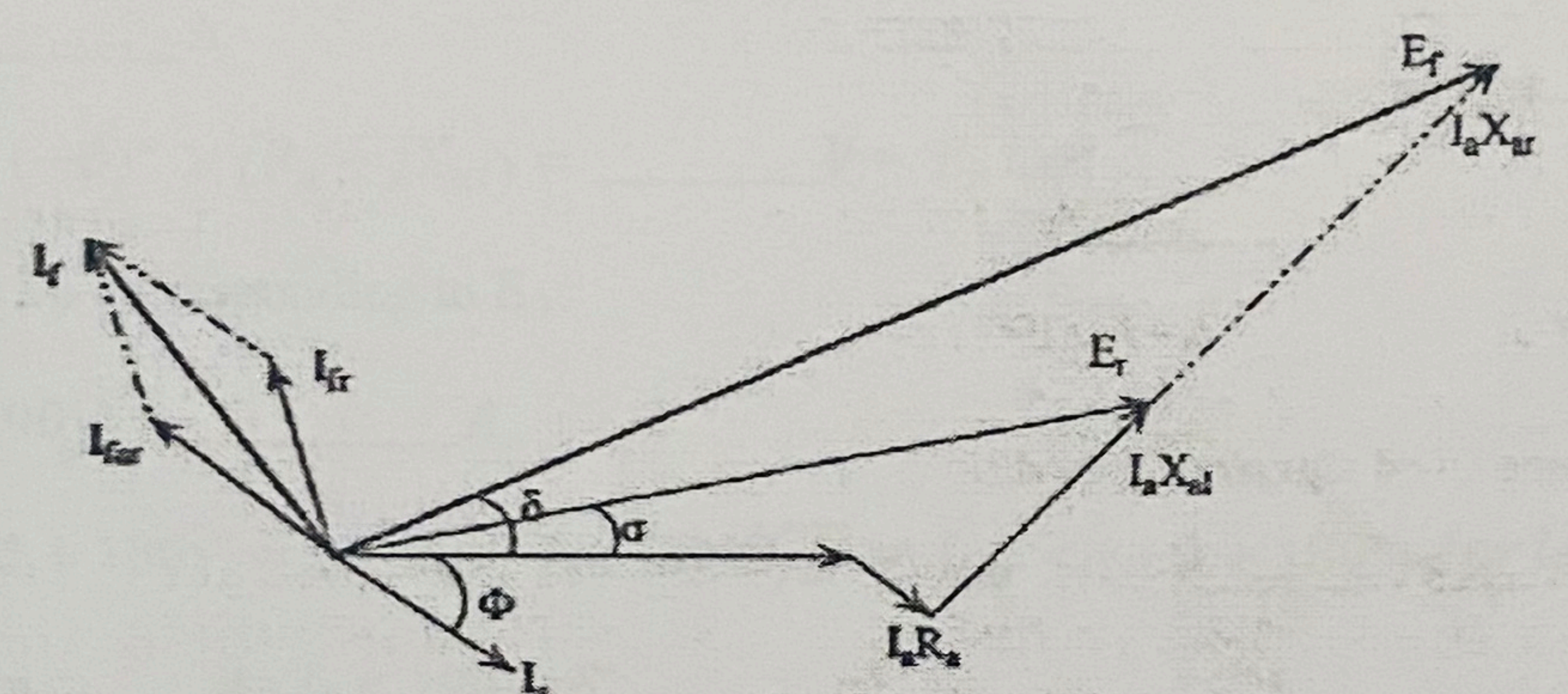
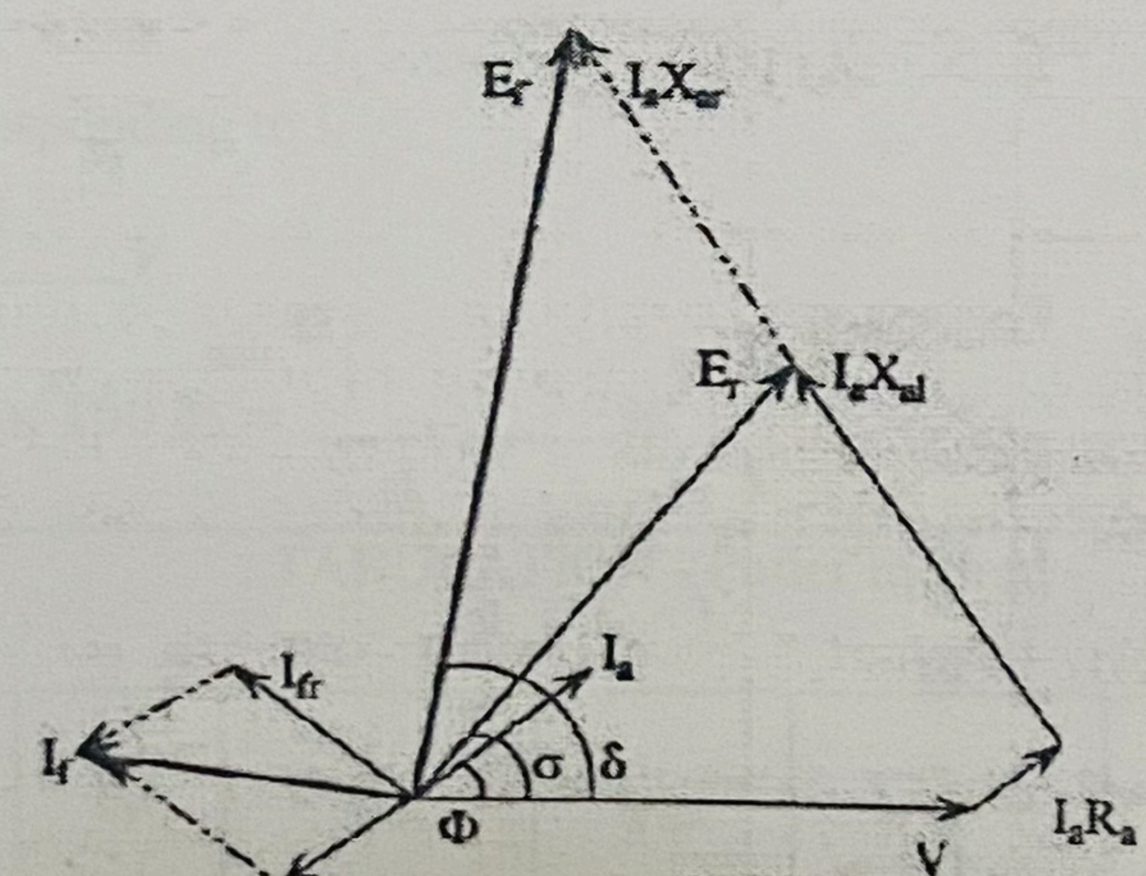
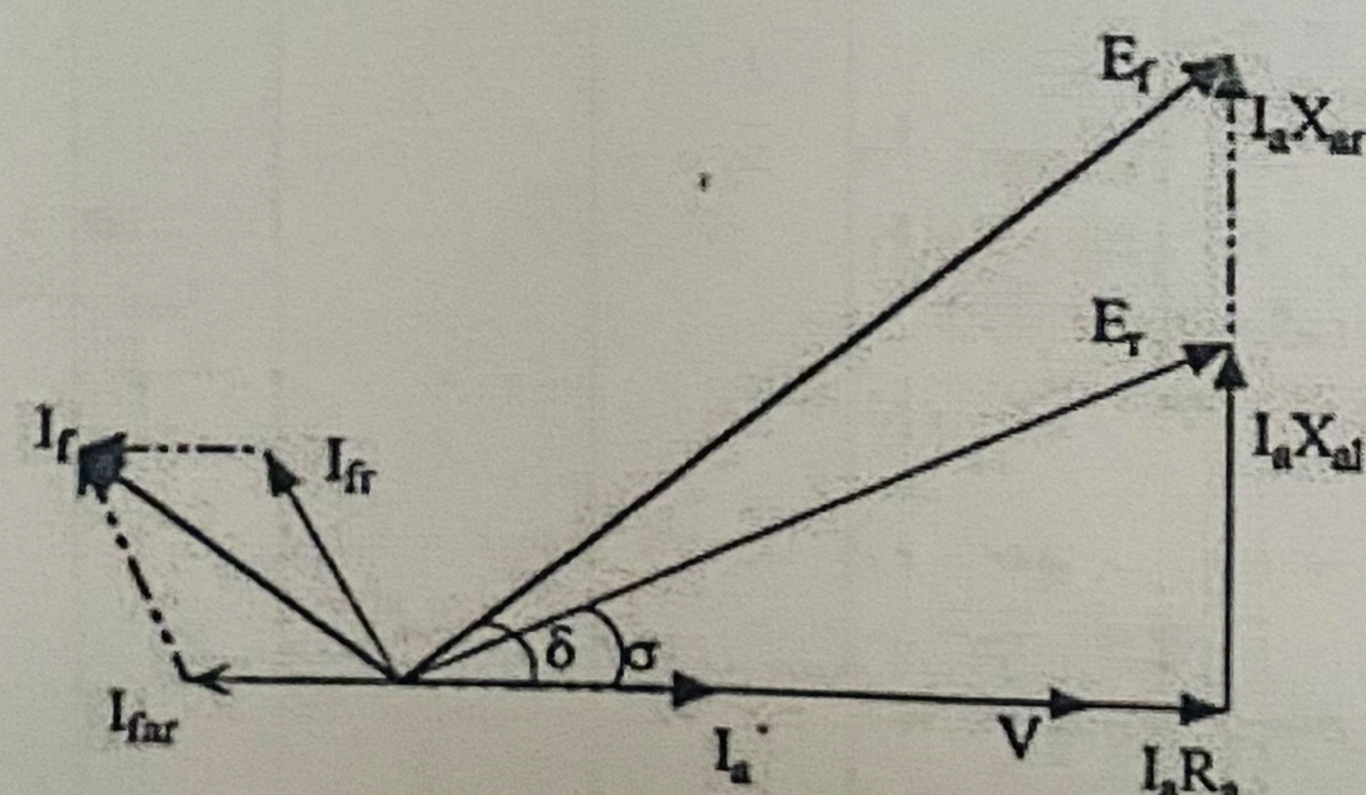
1. Draw AD = OF (parallel to X-axis)
2. Draw line DC from D parallel to air gap line meeting OCC at C.
3. Drop a vertical CB from C meeting the line AD at B.
4. Now, Δ ABC is the potier triangle.
5. Drop more potier triangles to complete ZPFC.
6. BC= $I_a X_{al}$= voltage drop due to armature leakage reactance
7. AB = I_{far} = Field current necessary to overcome the demagnetizing effect of armature reaction at full load.
8. BD = Field current necessary to induce an emf required for balancing leakage reactance drop BC.



Lagging power factor



Leading power factor



Unity power factor

SAMPLE CALCULATION